

LEXI: Priority-First Service Orchestration

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Abstract. A service spanning cloud, edge, and serverless nodes must balance latency, data exposure, carbon, and cost, which operators may rank strictly, e.g., meeting a latency SLO first. Weighted sums approximate such a ranking only with deployment-specific weights, and their decisions change when an objective’s numerical scale is misestimated. We present LEXI, a continuum orchestrator applying a thresholded lexicographic rule to predictions of an off-policy counterfactual outcome model, which assumes objectives decompose across stages. We instantiate the rule as $\text{SLO} \succ \text{privacy} \succ \text{carbon} \succ \text{cost}$ and use a simple topological privacy proxy only to separate placements already tied on the SLO. In a 15-seed simulation, LEXI needs no weights and reaches **1.46%** SLO violations with PII exposure **0.05**, comparable to hand-calibrated and SLO-constrained weighted sums. An untuned sum reaches 23% SLO violations and 1.94 PII exposure, and only one of 84 tested weightings meets a modest target. Under-scaling privacy $10\times$ raises the tuned and constrained baselines’ exposure to **1.85** and **1.84** of two stages, respectively. LEXI is unchanged. The evidence covers one simulated application under the additive no-interference assumption.

Keywords: Service orchestration · Cloud-edge-serverless continuum · Lexicographic optimisation · Scale invariance.

1 Introduction

Modern services run across a *continuum* [3] of regional cloud, on-premises edge, and serverless resources, where placement decides each stage’s latency, data exposure, carbon, and cost. Operators often rank these objectives, for example by requiring the latency service-level objective (SLO) to be met before they consider privacy, carbon, or cost. A weighted sum encodes such a ranking as exchange

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Artifact and extended version: <https://github.com/lexuanbach/lexi>.

rates between objectives. Widely separated weights can approximate it, but suitable values occupy a small, deployment-specific region (one of the 84 weightings tested here) and depend on units that change with hardware, carbon intensity, and budgets. Lexicographic selection needs no exchange rates and is invariant to strictly increasing per-objective transformations (Sect. 4).

We design LEXI to rank candidate placements on the highest-priority objective and break ties with lower priorities. This *thresholded lexicographic* rule (Fig. 1) reads an off-policy *counterfactual outcome model* that estimates untaken placements from executed ones. The evaluated controller makes no deliberate exploratory migration, though estimates beyond the observed assignments depend on model adequacy and feature coverage [4,14]. We instantiate the rule on SLO $\succ$ privacy $\succ$ carbon $\succ$ cost, where privacy is *the number of stages handling personally identifiable information (PII) on a publicly reachable node*. This deliberately coarse *topological* proxy is no compliance certificate and misses the misconfiguration failures dominating real exposure [6]. Ranked second, it never overrides the SLO and only separates SLO-tied candidates (Sect. 2), and another second-priority objective could replace it with the rule unchanged.

We simulate one application (Sect. 5) and make no physical-testbed claim. The extended version adds proofs, the input-generation derivation, a real-trace rerun and RQ5–RQ7 (slack, stability, safety margins, coverage probes, scaling).

2 The LEXI Model and Controller

Definition 1 (Continuum, service DAG, placement, objectives). A continuum is a set of nodes $\mathcal{N}$, each with a tier, a region, a privacy tier (edge private, others public) and the parameters of Sect. 5.1. A service DAG has stages $1, \dots, K$, each with a demand and a PII tag, and a placement π maps stages to nodes, giving for request r the vector $\mathbf{g}^{\text{raw}}(\pi, r) \in \mathbb{R}^4$ of critical-path latency, privacy (PII-tagged stages on a public node), carbon and cost, all lower is better.

Normalisation. Objective 0 is the SLO hinge $\mathbf{g}_0 = \min(1, \max(0, \text{latency} - \text{SLO})/\rho)$, $\rho = 60$ ms here. It treats latency as a threshold, which ties every SLO-feasible placement at $\mathbf{g}_0 = 0$ and every latency at or above 320 ms at $\mathbf{g}_0 = 1$. The other three are divided by an operator-declared magnitude and clamped to $[0, 1]$. While no candidate saturates the (only non-decreasing) clamp, changing those divisors is a strictly increasing reparameterisation, which leaves strict LEXI selection unchanged (Prop. 1), unlike a weighted sum.

Definition 2 (Priority order and selection rule). The operator specifies a strict priority order on the normalised coordinates, here SLO $\succ$ privacy $\succ$ carbon $\succ$ cost. The thresholded-lexicographic optimum with slack $\boldsymbol{\eta}$ keeps, for $k = 0, 1, 2$ in turn, the candidates within η_k of the current best on objective k , then breaks the final tie on cost. With $\boldsymbol{\eta} = \mathbf{0}$ it is the strict lexicographic optimum, consuming only the order of each objective’s values. With positive slack, additive distances and their units matter, and a lower priority may choose within the

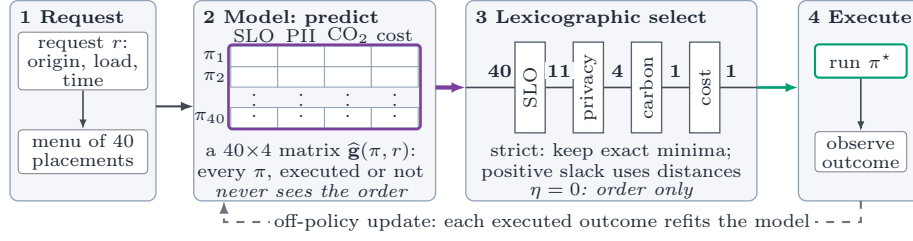


Fig. 1. LEXI’s decision loop: the model predicts every objective for every candidate (purple: the only interface). The evaluated strict rule ($\eta = 0$) reads order only (counts: Table 1).

tolerated higher-priority band. Under the strict rule, objective k is consulted only on the set objectives $0, \dots, k-1$ failed to separate, and a lower priority cannot cause a higher-priority regression.

The rule is a filter. Each level keeps the (near-)best on one column and passes them down. Per request LEXI predicts $\hat{\mathbf{g}}(\pi, r_t)$ for every candidate, applies Def. 2, executes and updates the model in $O(K)$. We evaluate the strict rule, $\eta = 0$, since slack on a threshold objective becomes SLO violations directly. Carbon slack, SLO safety margin and coverage probe (the extended version) are disabled.

Privacy coordinate and scope. $\mathbf{g}_1$ is a *topological* count, read only from the placement and a static sensitivity tag. Two of six stages carry the tag, putting “PII exposure” in $[0, 2]$. Ranked second, $\mathbf{g}_1$ is consulted only among SLO-tied placements (Sect. 3). It is no compliance verdict and ignores residency, encryption, transit security and the *misconfiguration* failures that dominate real exposure [6]. Hard prohibitions belong in the candidate generator or a residency pre-filter. $\mathbf{g}_1$ is a *preference* among otherwise-acceptable placements, *integer-valued* (hence tie-prone) and *unit-ambiguous*, and results resting only on these properties hold for any such objective.

Outcome-model interface. Under Assumption 1 LEXI keeps one online per-node estimator per objective, with error ε_t over the candidate menu. Model and rule share only a $|\mathcal{C}| \times 4$ matrix of normalised predictions. The model never learns the priority order, and the strict selector sees only each column’s ordering, without uncertainty or units. The error *can* still let a truly SLO-violating placement survive level 0, a band crossing that Thm. 1 bounds through ε_t .

3 A Worked Decision: From Signals to a Placement

The *relationship between the prediction model and the selection rule* is clearest on a real decision. Table 1 traces step $t = 904$ of seed 0 verbatim (node ids as in Table 2), and its “true” latency is for exposition only. From three exogenous signals (origin region 2, load $\ell = 0.765$, time $t = 904$), the model returns $\hat{\mathbf{g}}(\pi, r_{904})$

Table 1. Step $t=904$ (seed 0): six of the eleven level-0 survivors, including all four level-1 survivors. **Bold:** the choice. †: admitted at level 0 by prediction error.

#	Placement	Latency (ms)		Normalised $\hat{\mathbf{g}}$ (what the selector reads)				Elim.
		pred.	true	$\mathbf{g}_0$	$\mathbf{g}_1$	$\mathbf{g}_2$	$\mathbf{g}_3$	
2	E2/E2/E2/E2/E2/E2	252.5	251.9	0	0.0	0.673	1.000	L2
6	S2/S2/S2/S2/S2/S2	228.9	228.8	0	1.0	0.166	0.216	L1
19	C2/E2/C2/C2/C2/E2	230.3	245.8	0	0.0	0.476	0.848	L2
20	S1/E2/S1/S1/S1/E2	259.1	259.0	0	0.0	0.387	0.608	L2
21	S2/E2/S2/S2/S2/E2	235.7	235.5	0	0.0	0.314	0.592	—
24 [†]	C0/C2/S2/S1/C2/C0	256.2	268.2	0	1.0	0.371	0.496	L1

for *all* 40 candidates, including the 39 not executed, via per-node models fitted on steps 0–903. This 40×4 matrix is *all* the selector sees.

Level 0, the SLO hinge: $40 \rightarrow 11$. Every predicted-feasible placement scores 0, which leaves 11 of 40 candidates tied at the predicted minimum $\mathbf{g}_0 = 0$ for the next priority to break. Only 9 of the 11 truly meet the 260ms SLO, as under-prediction admitted #24 and one other (not shown). This is how estimation error reaches the top priority (ε_t in Thm. 1), and both fall at level 1.

Level 1, privacy: $11 \rightarrow 4$. Of the eleven, the minimum predicted exposure is $\mathbf{g}_1 = 0$, and the four with both PII stages on a private edge node survive, {2, 19, 20, 21}: **privacy is consulted only on the set the SLO has already tied, and can only ever choose among SLO-equivalent placements.**

Levels 2–3. #21 wins the near-continuous carbon level outright, leaving the cost tie-break unused. A band $\eta_2=0.08$ would keep #20 as a near-tie ($0.387 \leq 0.314 + 0.08$) and defer to cost (again #21). On the *same* matrix the untuned sum picks #6, exposing **both** PII stages, because #6 wins on carbon and cost and the sum prices two exposed stages below that saving. That exchange rate *is* the weighting. The tuned sum (0.030 vs. 0.111 for #6) and Constr-WS pick #21, yet a $10\times$ privacy under-scaling switches the tuned sum to #6 (0.021 vs. 0.030). No such rescaling changes the lexicographic choice (Prop. 1).

4 Theoretical Analysis

Full proofs are in the extended version. Both results hold for any ordered bundle of bounded objectives, whatever the coordinates mean. A decision costs $O(|\mathcal{C}|K)$.

Proposition 1 (Scale-invariance). *Let $\phi = (\phi_0, \dots, \phi_3)$ be any strictly increasing per-objective reparameterisations, and replace each $\mathbf{g}_k$ by $\phi_k \circ \mathbf{g}_k$ in the estimates the selector sees. Strict lexicographic selection returns the same placement. A fixed-weight scalariser is not invariant in general: there exist candidate sets and positive coordinate rescalings that change its selected optimum. A pair’s order flips only when the scaled-coordinate difference and weighted remainder have opposite signs. Another candidate may still preserve the global argmin.*

Such invariance is classical [8]. Pricing makes any method *scale-dependent*, including a constrained formulation beyond the coordinate it constrains (Sect. 6.2).

Assumption 1 (Additive no-interference / stable unit treatment value assumption (SUTVA)). *Each objective decomposes as a sum (latency: critical path) of per-(stage, node) terms that depend on the stage’s assignment and the exogenous request state and not on which other stages are co-placed. Executed placements update their per-node terms. Prediction elsewhere also requires model adequacy and node-feature coverage.*

Co-location contention breaks Assumption 1 for every additive-model method, and Sect. 6.3 finds a graded degradation largely preserving the ordering. Real shared-node interference is untested.

Theorem 1 (Top-priority tracking, under assumptions). *Let π_t^* be the priority-optimal placement in menu $\mathcal{C}_t$, and let $\varepsilon_t = \max_{\pi \in \mathcal{C}_{t,k}} |\hat{\mathbf{g}}_{t,k}(\pi) - \mathbf{g}_{t,k}(\pi)|$. Then $\mathbf{g}_0(\hat{\pi}_t) \leq \mathbf{g}_0(\pi_t^*) + \eta_0 + 2\varepsilon_t$. If also (A1) the per-node models are realisable up to a floor ε_{mis} , (A2) features and parameters are bounded, noise sub-Gaussian, and (A3) every per-node Gram matrix is persistently excited ($\lambda_{\min} \geq ct$), then with probability $1 - \delta$ online ridge estimation [1] bounds the cumulative top-priority gap by $\eta_0 T + O(\sqrt{T \log T}) + 2\varepsilon_{\text{mis}} T$. Because a no-exploration policy does not guarantee (A3), that rate is conditional. The bound is on normalised hinge loss and covers neither raw latency nor the SLO-violation probability.*

5 Experimental Setup

We use a purpose-built discrete-event simulator (released with the artifact) with $M=7$ nodes over 3 regions, a $K=6$ -stage DAG with a 260 ms SLO, and a geo-diurnal workload varying load and origin. Every method ranks the same fixed 40-candidate menu. The primary comparison uses synthetic load and carbon signals and averages **15 seed-level post-warm-up summaries** over steps 800–1599. 95% bootstrap intervals and paired tests resample those summaries. A separate rerun, without re-tuning, substitutes UK Carbon Intensity API forecasts [16] and the Azure Functions 2019 trace [17] for those two signals.

5.1 How the evaluation inputs are generated, and why

Energy, cost, round-trip time (RTT), queueing, and PII tags are authorial choices: DeathStarBench [10] and Alibaba provide DAG shapes but no continuum placements. Table 2 deliberately creates tension among private but weak edge nodes, powerful remote cloud nodes, and cheap public serverless nodes [3]. Values are simulator units, with no provider measurements behind them. The synthetic intensity ordering prevents privacy and carbon from aligning trivially.

The DAG, the PII tags, and the menu. The evaluated DAG is `frontend(8) → auth(16) → {search(30), recommend(25)} → order(20) → pay(18)`, with per-stage compute demand in parentheses. Stage s on node n has latency $\text{RTT}(\text{origin}, n) + b_n \text{demand}_s + q_n \ell$, with RTT_{in} within the node’s region and RTT_{out} across, b the compute per unit demand (as is energy) and q the queueing

Table 2. Node parameters, verbatim from the artifact. Edge nodes are private.

Id	Tier	Reg.	Latency terms				Cost and carbon			
			b	RTT _{in}	RTT _{out}	q	Cost	Energy	$\bar{I}$	Priv.
			ms/unit	ms	ms	ms/load	sim./inv.	Wh, $d=20$	g/kWh	
E0	edge	0	2.20	4	70	9.0	1.15	0.62	520	✓
E1		1	2.00	5	72	8.5	1.10	0.58	540	✓
E2		2	2.10	5	71	8.8	1.12	0.60	430	✓
C0	cloud	0	0.50	35	46	2.0	0.55	0.95	360	
C2		2	0.55	38	50	2.2	0.50	0.90	190	
S1	serverless	1	1.10	18	30	4.5	0.20	0.36	260	
S2		2	1.20	20	32	4.8	0.18	0.34	175	

sensitivity per unit load. Its carbon (g) is $\sum_s \text{energy}_n(\text{demand}_s/20)I_n(t)/1000$, where I_n is a synthetic node-specific effective intensity oscillating around $\bar{I}_n$ with an amplitude of 60–120 gCO₂/kWh on a 240-step diurnal cycle, and queueing and carbon noise are seeded at execution. The DAG’s shape (serial head, compute-heavy parallel fan-out, join, serial tail) follows DeathStarBench [10] and the Alibaba traces [13]. We set demands so that the parallel branches dominate the critical path. The two PII-tagged stages, **auth** and **pay**, also contribute latency while determining privacy. Thus the objectives are coupled, and $\mathbf{g}_1$ is integer-valued and tie-prone. The queueing sensitivities q have the least external grounding. We set them proportional to inverse compute capacity (the weak edge queues hardest) and large enough that load re-orders the tiers over a day, and the SLO sweep of Sect. 6.3 serves as the sensitivity analysis for this choice. The deduplicated menu holds 22 *structured* placements and 18 seeded random fills. The structured ones are 7 homogeneous placements (one cost-minimal), 3 latency-greedy ones (one per origin region), and 12 privacy-aware ones enumerating every (private edge, public backend) pair. These last guarantee that the privacy objective always has something to choose, which pins any privacy failure on the *rule*. The menu is not neutral, however, and Sect. 6.3 shows that it is the largest determinant of the absolute numbers.

Baselines. All methods share the menu, model architecture, and initial priors. Each learned method maintains its own model and therefore acquires a policy-dependent training trajectory. LEXI has no weights, and the weight-based baselines get favourable calibrated settings: *WS-default* (untuned 0.30, 0.12, 0.30, 0.28), *WS-tuned* (calibrated, SLO-heavy 0.85, 0.10, 0.00, 0.05, better than the best of the 84-point sweep), three *scale-robust* scalarisers reusing those weights [15], *Lex-LP*, and *Constr-WS*. Constr-WS, a one-step selector with no learned policy, admits only predicted-SLO-feasible candidates and then minimises an equal-weight sum of the rest. If none is predicted feasible, it chooses the candidate with minimum predicted SLO hinge. Lex-LP, the same rule without LEXI’s hold (keeping a previous placement that survives levels 0–2), matches LEXI on all 24,000 decisions. We also report the *priority-inversion rate*, the share

Table 3. RQ2 operator-facing outcomes (15 seeds, means, lower is better). Inv%: priority-inversion rate. Bold: best or tied. Scale-robust sums: see text.

Method	SLO%↓	PII↓	Carbon↓	Cost↓	Churn↓	Inv%↓
LEXI (no weights)	1.46	0.05	0.98	3.21	1.35	0.0
Constr-WS (SLO filter + sum)	1.32	0.05	0.98	3.19	1.32	0.7
WS-tuned (calibrated)	3.19	0.01	0.99	3.20	1.28	3.7
WS-default (untuned)	23.02	1.94	0.45	1.22	0.65	94.7

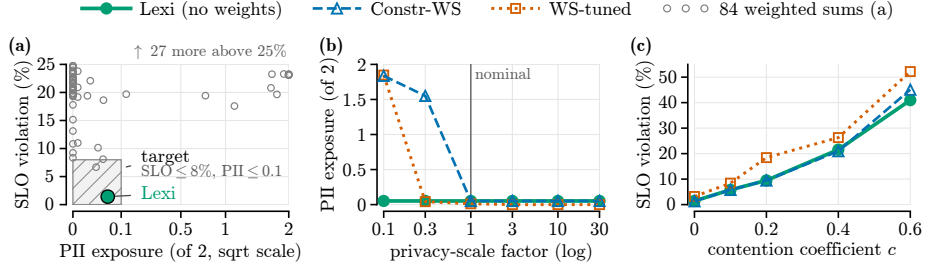


Fig. 2. (a) RQ1 weight sweep, hatched: target (6 seeds). (b) RQ3 privacy-scale miscalibration (8 seeds). (c) RQ4 co-location contention, breaking Assumption 1 (15 seeds).

of decisions lexicographically worse than the strict priority optimum on shared estimates. Churn counts the stages moved per request.

6 Results

6.1 RQ1/RQ2: tuning and objective satisfaction

A weighted sum reaches the paper’s target operating region only after a deployment-specific search. Of 84 weightings swept over the objective simplex (Fig. 2(a)), only **one** meets a modest target ($\text{SLO} \leq 8\%$ and $\text{PII} \leq 0.1$) and **none** dominate LEXI (1.42% SLO in this 6-seed sweep). In the 15-seed main run (Table 3), LEXI has **1.46%** SLO violations (95% CI [1.31, 1.60]) at PII exposure **0.05** from the order alone. It is on par with Constr-WS and better on SLO than the hand-tuned sum (paired $p < 0.001$), which is better on privacy. These methods occupy a nearby target region with different residual trade-offs. LEXI needs no weight tuning. Scale-robust sums approach LEXI’s privacy at 2.3–3.3 $\times$ its SLO violation, and only LEXI never inverts the order (0.0% vs. 0.7–94.7%).

6.2 RQ3: sensitivity to objective scaling

We next rescale the *units* of the privacy objective as the optimiser sees them, leaving true outcomes unchanged. LEXI’s PII exposure stays **invariant** at 0.052 across a 300 $\times$ swing (Fig. 2(b)), because selection depends only on the objective *order* (Prop. 1). The tuned sum matches LEXI’s privacy at 2.3–3.3 $\times$ its SLO violation. Under-weighting privacy 10 $\times$ drives its PII from 0.01 to **1.85 of 2**

stages (augmented Tchebycheff: 1.86), and such an operator-declared scale can drift. The rank- and min-max-normalised sums are invariant too, yet they sit at $\approx 3.5\%$ SLO because they still *price* what they rank.

Comparison with Constr-WS. Constr-WS also needs no tuning for its SLO constraint and is close to LEXI on the headline row (1.32% vs. 1.46%, with identical PII and carbon, and cost within 0.02). On the top priority the two use the same feasible set when at least one candidate is predicted SLO-feasible, because the hard constraint is the strict $\eta_0=0$ top-level band. Below the filter Constr-WS applies a price where LEXI applies three more levels. Constr-WS inverts the priority order on 0.68% of decisions here and on **14.3%** under the real Azure load (the extended version), against exactly 0% for LEXI in both, and a $10\times$ privacy under-scaling raises its exposure from 0.052 to **1.84 of 2 stages** (Fig. 2(b)). Its hard SLO constraint is scale-free, but the blend beneath it is not.

6.3 RQ4: robustness to curation, configuration and additivity

Curation matters most: a random menu moves LEXI from 1.44% to **43.6%** SLO violations (8 seeds). At a typical request 13.7% of the 40 curated candidates meet the SLO, 45.0% keep both PII stages private and 5.6% do both, and in the random menu these shares collapse to 3.9%, 19.4% and **0.2%**. The *expected-objective oracle* selects the priority optimum from noise-free simulator expectations and is then scored on a realised outcome. It violates the SLO on 1.48% of requests with the curated menu vs. LEXI’s 1.44%. This ordering is possible because realised draws differ. **With the random menu the oracle itself violates on 43.53%** vs. LEXI’s 43.56%. Thus absolute outcomes depend on the menu, while method ordering survives (43.6% LEXI, 45.8% WS-tuned, 59.8% WS-default).

The ordering also holds over an SLO sweep of 220–340 ms and an independent 10-node, 8-stage configuration with three PII stages (LEXI 6.78% < Constr-WS 7.17% < WS-tuned 8.14% < WS-default 13.81%). Co-location contention breaks Assumption 1 (Fig. 2(c)) and degrades *every* additive-model method together and gradually, preserving the ordering of Table 3 until $c=0.6$, where Constr-WS falls behind LEXI. It damages both SLO and privacy: LEXI’s SLO violation rises from 1.46% to 40.97% and PII exposure from 0.05 to 1.49. Over the *whole* menu, including the 39 unexecuted candidates per step, the model’s mean absolute SLO-hinge error is **0.0286** normalised, and LEXI selects the expected-objective priority optimum on **99.7%** of decisions at under 0.1 ms each.

7 Discussion and Threats to Validity

We deliberately create priority conflict in the simulator (Sect. 5.1). The candidate menu determines the achievable absolute outcomes (Sect. 6.3), while additivity (Assumption 1) defines the model boundary. The evaluation isolates the decision rule from control-plane overhead, migration cost, telemetry delay, admission control, cold starts, and uninjected failures. It covers one application,

and the privacy coordinate is a ranked topological preference and no compliance model. Hardware validation with concurrent applications is the next step from decision-level evidence to deployment performance.

8 Related Work

In continuum scheduling [3], three recent systems are closest. *HuntMS* [5] geo-distributes microservices for carbon and cost under an SLO *constraint*, *Cluster-Less* [7] orchestrates serverless workflows against a *deadline*, and *Festina* [18] minimises serverless energy under a latency-SLO constraint. Each optimises one scalar under a constraint without ranking the rest. We compare formulations and do not run these systems: their substrates are absent from our simulator, while *Constr-WS* (Sect. 6.2) captures the shared decision pattern of filtering on a threshold objective and then scalarising the feasible set. This excludes their engineering and limits the comparison. Constrained MDPs [2] likewise impose “meet the SLO first” as a constraint (in expectation) and then price the rest. Early lexicographic RL [9] ranks objectives in tabular settings. Temporal-logic objectives also avoid hand-set weights and reduce optimally to average rewards [12], but constrain behaviour over time, whereas LEXI orders objectives in one decision. The same priority-first pattern governs post-quantum profile selection, with admissibility gates before cost ranking [11].

9 Conclusion

Operators express priorities as an order. We encode that order directly, without exchange rates. LEXI applies strict lexicographic selection to predictions from an off-policy counterfactual outcome model. In our simulation, it reaches the weighted and constrained baselines’ target region without tuning and remains unchanged under strictly increasing per-objective transformations. This result belongs to strict selection: positive slack introduces unit-dependent bands, and the candidate menu determines what any selector can achieve.

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